

Decision Table

1. Restaurant Table Reservation Confirmation

Decision Table Conditions

- C1: Party Size ≤ 8
- C2: Advance Payment Made
- C3: Peak Hour Slot

Decision Table (Rules)

Rule	C1	C2	C3	Action / Expected Result
R1	Y	Y	N	Reservation confirmed instantly
R2	Y	Y	Y	Reservation confirmed with peak-hour surcharge
R3	Y	N	N	Reservation held pending payment
R4	N	Y	N	Manager approval required – large party
R5	Y	Y	N	Reservation confirmed instantly (boundary)
R6	N	N	Y	Reservation rejected

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC1_01	Party = 4, Payment = Yes, Peak Hour = No	Reservation confirmed instantly	R1
TC1_02	Party = 4, Payment = Yes, Peak Hour = Yes	Reservation confirmed with peak-hour surcharge	R2
TC1_03	Party = 4, Payment = No, Peak Hour = No	Reservation held pending payment	R3
TC1_04	Party = 12, Payment = Yes, Peak Hour = No	Manager approval required – large party	R4
TC1_05	Party = 8, Payment = Yes, Peak Hour = No	Reservation confirmed instantly	R5
TC1_06	Party = 12, Payment = No, Peak Hour = Yes	Reservation rejected	R6

Individual Explanation of Each Test Case

TC1_01: Validates the standard happy path for a small, paid, off-peak reservation.

TC1_02: Verifies a peak-hour surcharge is added even when payment is already made.

TC1_03: Confirms an unpaid reservation is held rather than confirmed.

TC1_04: Ensures parties above the size limit require manager approval.

TC1_05: Tests the boundary value (exactly 8 guests) to confirm it is still auto-confirmed.

TC1_06: Verifies rejection when the party is oversized, unpaid, and during peak hours.

2. Password Reset Validation

Decision Table Conditions

- C1: Email Registered
- C2: OTP Correct
- C3: Reset Link Expired

Decision Table (Rules)

Rule	C1	C2	C3	Action / Expected Result
R1	Y	Y	N	Password reset successful
R2	Y	N	N	Invalid OTP message shown
R3	Y	Y	Y	Link expired message shown
R4	N	-	-	Email not found message shown
R5	Y	Y	N	Password reset successful (boundary)
R6	N	N	N	Email not found message shown

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC2_01	Email registered, OTP correct, Link not expired	Password reset successful	R1
TC2_02	Email registered, OTP incorrect, Link not expired	Invalid OTP message shown	R2
TC2_03	Email registered, OTP correct, Link expired	Link expired message shown	R3
TC2_04	Email not registered	Email not found message shown	R4
TC2_05	Email registered, OTP entered at last valid second, Link not expired	Password reset successful	R5
TC2_06	Email not registered, OTP incorrect, Link expired	Email not found message shown	R6

Individual Explanation of Each Test Case

TC2_01: Verifies a successful reset when all conditions are correct.

TC2_02: Ensures an incorrect OTP is caught before the reset proceeds.

TC2_03: Confirms an expired link blocks the reset even with a correct OTP.

TC2_04: Ensures an unregistered email is rejected immediately.

TC2_05: Tests the boundary case of the OTP being entered just before expiry.

TC2_06: Confirms the email check takes priority when multiple fields are invalid.

3. Elevator Overload Safety System

Decision Table Conditions

- C1: Weight $\leq$ Maximum Capacity
- C2: Door Fully Closed
- C3: Emergency Stop Pressed

Decision Table (Rules)

Rule	C1	C2	C3	Action / Expected Result
R1	Y	Y	N	Elevator moves normally
R2	N	Y	N	Overload alarm sounds – elevator halted
R3	Y	N	N	Doors remain open – elevator waits
R4	Y	Y	Y	Emergency stop – elevator halted
R5	Y	Y	N	Elevator moves normally (boundary)
R6	N	N	Y	Overload alarm and emergency stop – elevator halted

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC3_01	Weight = under capacity, Door = closed, Emergency Stop = No	Elevator moves normally	R1
TC3_02	Weight = over capacity, Door = closed, Emergency Stop = No	Overload alarm sounds – elevator halted	R2
TC3_03	Weight = under capacity, Door = open, Emergency Stop = No	Doors remain open – elevator waits	R3
TC3_04	Weight = under capacity, Door = closed, Emergency Stop = Yes	Emergency stop – elevator halted	R4
TC3_05	Weight = exactly at capacity, Door = closed, Emergency Stop = No	Elevator moves normally	R5
TC3_06	Weight = over capacity, Door = open, Emergency Stop = Yes	Overload alarm and emergency stop – elevator halted	R6

Individual Explanation of Each Test Case

TC3_01: Validates normal operation when weight and door conditions are both satisfied.

TC3_02: Ensures the overload alarm triggers when capacity is exceeded.

TC3_03: Confirms the elevator will not move while the doors are open.

TC3_04: Ensures the emergency stop halts the elevator regardless of other conditions.

TC3_05: Tests the boundary value (exactly at maximum capacity) to confirm normal operation.

TC3_06: Verifies that overload and emergency stop conditions can be reported together.

4. Insurance Claim Approval

Decision Table Conditions

- C1: Policy Active

- C2: Claim Amount $\leq$ Coverage Limit
- C3: Documents Verified

Decision Table (Rules)

Rule	C1	C2	C3	Action / Expected Result
R1	Y	Y	Y	Claim approved
R2	Y	Y	N	Claim pending – documents under verification
R3	Y	N	Y	Claim partially approved, up to coverage limit
R4	N	-	-	Claim rejected – policy inactive
R5	Y	Y	Y	Claim approved (boundary)
R6	N	N	N	Claim rejected

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC4_01	Policy active, Claim = ₹40,000 (within limit), Documents verified	Claim approved	R1
TC4_02	Policy active, Claim = ₹40,000, Documents not verified	Claim pending – documents under verification	R2
TC4_03	Policy active, Claim = ₹1,20,000 (exceeds limit), Documents verified	Claim partially approved, up to coverage limit	R3
TC4_04	Policy inactive	Claim rejected – policy inactive	R4
TC4_05	Policy active, Claim = exactly the coverage limit, Documents verified	Claim approved	R5
TC4_06	Policy inactive, Claim exceeds limit, Documents not verified	Claim rejected	R6

Individual Explanation of Each Test Case

TC4_01: Validates full approval when the policy, amount, and documents all meet requirements.

TC4_02: Confirms a claim stays pending until documents are verified.

TC4_03: Ensures claims above the coverage limit are only approved up to that limit.

TC4_04: Verifies claims are rejected outright when the policy is not active.

TC4_05: Tests the boundary value where the claim exactly equals the coverage limit.

TC4_06: Confirms rejection when none of the required conditions are met.

5. Flight Ticket Refund Policy

Decision Table Conditions

- C1: Cancelled $\geq$ 24 Hours Before Departure
- C2: Refundable Ticket Type

- C3: Airline-Initiated Schedule Change

Decision Table (Rules)

Rule	C1	C2	C3	Action / Expected Result
R1	Y	Y	N	Full refund issued
R2	Y	N	N	No refund – non-refundable ticket
R3	N	Y	N	Partial refund – cancellation fee applies
R4	-	-	Y	Full refund issued – airline-initiated change
R5	Y	Y	N	Full refund issued (boundary)
R6	N	N	N	No refund

Test Cases Derived from the Table

TC ID	Test Scenario / Inputs	Expected Output	Rule Covered
TC5_01	Cancelled 48 hrs before, Refundable ticket, No schedule change	Full refund issued	R1
TC5_02	Cancelled 48 hrs before, Non-refundable ticket, No schedule change	No refund – non-refundable ticket	R2
TC5_03	Cancelled 10 hrs before, Refundable ticket, No schedule change	Partial refund – cancellation fee applies	R3
TC5_04	Any cancellation timing, Non-refundable ticket, Airline schedule change = Yes	Full refund issued – airline-initiated change	R4
TC5_05	Cancelled exactly 24 hrs before, Refundable ticket, No schedule change	Full refund issued	R5
TC5_06	Cancelled 10 hrs before, Non-refundable ticket, No schedule change	No refund	R6

Individual Explanation of Each Test Case

TC5_01: Validates a full refund when cancellation happens well before departure on a refundable ticket.

TC5_02: Confirms non-refundable tickets receive no refund even with early cancellation.

TC5_03: Ensures late cancellations on refundable tickets still incur a cancellation fee.

TC5_04: Verifies an airline-initiated schedule change guarantees a full refund regardless of ticket type.

TC5_05: Tests the boundary value (exactly 24 hours before departure) to confirm full refund eligibility.

TC5_06: Confirms no refund is issued when none of the favorable conditions are met.